

Kalpana Hackathon- 2025

Create, Collaborate, Conquer

Hack for Humanity

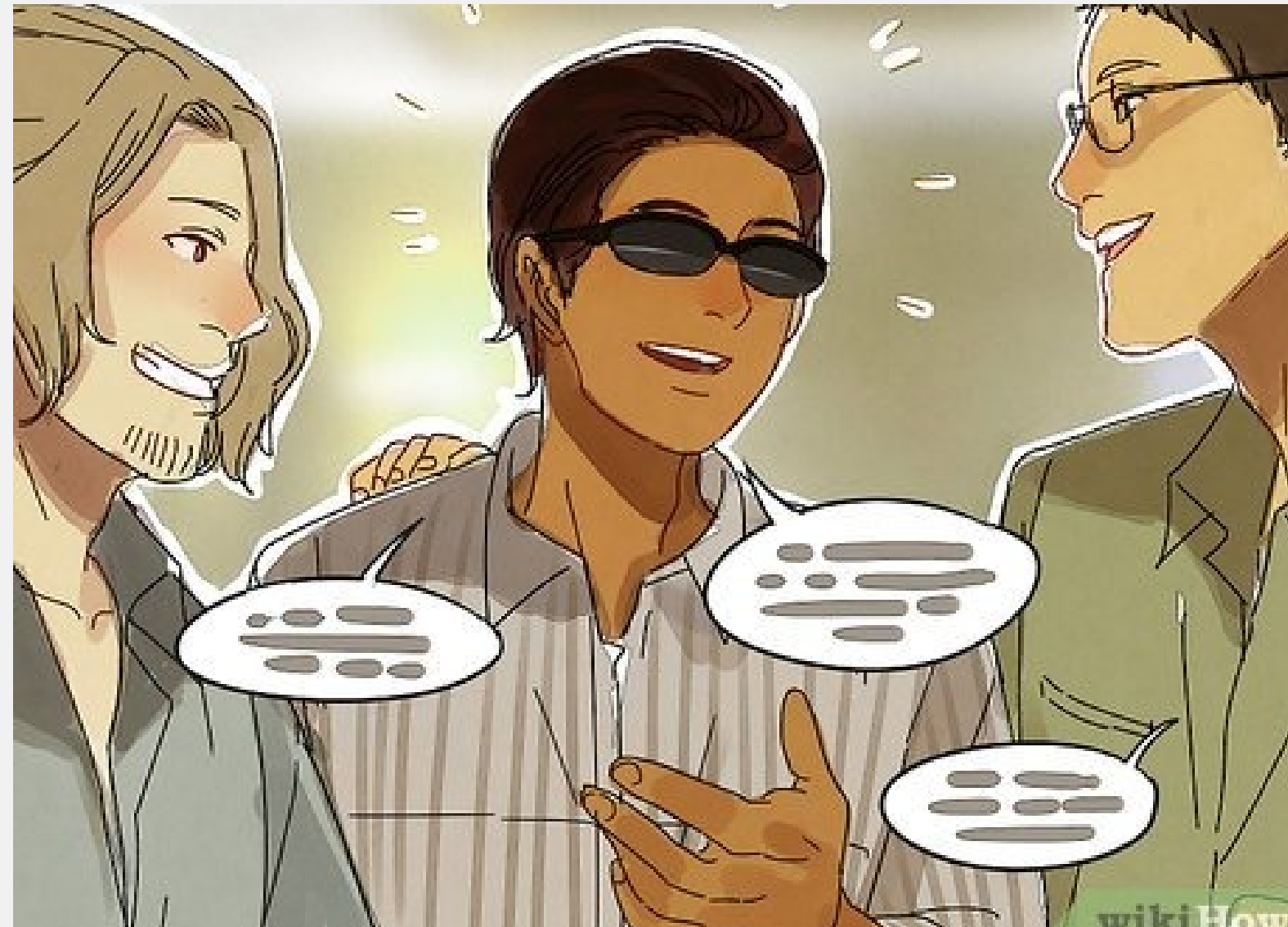
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1st Year



Problem Statement

The program removes communication gap between blind and deaf/mute individuals by enabling **speech-to-text** and **text-to-speech** conversion



Introduction

India has an estimated 4.95 million **blind** people, which is about **0.36%** of the total population and approximately 63million people are have significant **hearing** impairment which is **6.3%** of the population.

Keeping this in mind we are helping people who are blind, deaf and mute. We are choosing **AI/ML track** for this using text-to-speech-conversion and speech-to-text conversion.



Workflow

1.Start GUI:

The system launches a welcome message and a "Start Conversation" button.

2. Deaf/Mute user :

Types a message, which is converted to speech for the blind user

3.Blind User:

Speaks and the system converts speech to text for the deaf/mute user.

4.Face Detection:

Activates speech recognition when a face is detected.

5.Real-Time Conversation:

Messages appear in a shared chat window with reset options

Requirements

Software Requirements

1. Python 3.x – Programming language
2. Tkinter – GUI development
3. Pytttsx3 – Text-to-speech conversion
4. SpeechRecognition – Speech-to-text processing
5. OpenCV – Face detection using webcam
6. Pillow (PIL) – Image handling for GUI

Requirements

Hardware Stack:

1.Keyboard

2.Camera

3.Microphone

4.Speaker

Use cases of this program

1. **Personal and Social Interaction** between blind and deaf/mute individuals.

2. **Emergency situations** where accessibility tools are needed.

3. **Education** for visually or hearing-impaired students.

4. **Customer Support and Service**

5. **Assistive Communication for Elderly**

6. **Deaf /mute travellers**



Advantages

1. Enhanced communication
2. Flexibility and Customisation
3. Increase confidence
4. Improve quality of life
5. Accessibility and easy to use



Conclusion

By bridging the communication gap, this code has the potential to positively impact the lives of millions of people worldwide and promoting a more inclusive, equitable and compassionate world for all.



THANK

YOU !